

to copying it. It should come as no surprise, then, that the standard GNU `coreutils` `cp` command comes with a `-l` flag that causes it to create (hard) links instead of copies (it doesn't hard-link directories, though, which is good; you might want to think about why that is). Another handy switch for the `cp` command is `-a` (archive), which causes it to recurse through directories and preserve file owners, timestamps, and access permissions.

Together, the combination `cp -al` makes what appears to be a full copy of a directory tree, but is really just an illusion that takes almost no space. If we restrict operations on the copy to adding or removing (unlinking) files – i.e., never changing one in place – then the illusion of a full copy is complete. To the end-user, the only differences are that the illusion-copy takes almost no disk space and almost no time to generate.

2002.05.15: Portability tip: If you don't have GNU `cp` installed (if you're using a different flavour of *nix, for example), you can use `find` and `cpio` instead. Simply replace `cp -al a b` with `cd a && find . -print | cpio -dpl ../b`. Thanks to Brage Førlund for that tip.

10.8 Putting it all together

We can combine `rsync` and `cp -al` to create what appear to be multiple full backups of a file system without taking multiple disks' worth of space. Here's how, in a nutshell:

```
rm -rf backup.3
mv backup.2 backup.3
mv backup.1 backup.2
cp -al backup.0 backup.1
rsync -a --delete source_directory/ backup.0/
```

If the above commands are run once every day, then `backup.0`, `backup.1`, `backup.2`, and `backup.3` will appear to each be a full backup of `source_directory/` as it appeared today, yesterday, two days ago, and three days ago, respectively – complete, except that permissions and ownerships in old snapshots will get their most recent values (thanks to J.W. Schultz for pointing this out). In reality, the extra storage will be equal to the current size of `source_directory/` plus the total size of the changes over the last three days – exactly the same space that a full plus daily incremental backup with `dump` or `tar` would have taken.

Update (2003.04.23): As of `rsync-2.5.6`, the `-link-dest` flag is now standard. Instead of the separate `cp -al` and `rsync` lines above, you may now write:

```
mv backup.0 backup.1
rsync -a --delete --link-dest=../backup.1 source_
directory/ backup.0/
```

This method is preferred, since it preserves original permissions and ownerships in the backup. However, be sure to test it – as of this writing